BGIN Block #10 Meeting Report

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Layer2 Governance Roundtable

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Summary

Session Report: Layer Two Governance in Blockchain Technology

Date: 5th March 2024

Reference to Keynote:

BGIN Block #10 Meeting Report_Layer2 Governance_KeyNote	Discourse

Overview

During a recent roundtable, discussions were centered around governance challenges and opportunities within Layer Two blockchain solutions. Mitchell initiated the session by setting the tone for a collaborative and confidential dialogue under Chatham House rules. This approach was aimed at fostering open discussions without attributing specific insights to individual contributors, with a focus on leveraging insights provided by a contributor Justin Drake, an Ethereum Researcher.

Main Discussants

- Mitchell, Session Chair, IKP Working Group, Soulbis
- Leona, representing INTMAX, shared her project's focus on developing a ZKP Layer Two solution aimed at addressing transaction costs on Ethereum.
- Jack, a co-founder of Pier Two, elaborated on his company's extensive involvement in blockchain infrastructure since 2018, highlighting their activities in the Ethereum and Arbitrum ecosystems and their efforts in developing a light client in C sharp.

Key Discussion Themes

- **Governance Challenges in zk Layer 2 Solutions:** The complexity of zk proofs (ZKP) and the difficulty in auditing them were identified as significant governance and security challenges. Leona highlighted the potential of AI models, specifically LLMs, to automatically detect bugs, underscoring the need for innovative approaches to maintain system security.
- **Strategies for Enhancing Security:** The discussion underscored mainstream strategies like multi-signature and two-factor authentication methods, aiming to ensure the redundancy and integrity of Layer 2 solutions. This approach seeks to protect user assets by maintaining system reliability, even if one part of the system is compromised.
- **Adjusting Penalty Mechanisms:** Jack discussed the implications of capping slashing penalties for blockchain validators, emphasizing the potential benefits for staking service

providers and the broader ecosystem by simplifying risk assessment for insurance purposes.

- **The Future of Layer 2 Governance:** The conversation explored the potential for integrating zk EVM into Ethereum's Layer 1 to enhance scalability and verification processes. This sparked a broader discussion on the evolving role of Layer 2 solutions in the blockchain ecosystem and the importance of governance models that adapt to technological advancements.
- **Retroactive Public Goods Funding:** The concept of rewarding contributors based on past contributions rather than potential future returns was discussed as a strategy for supporting the development and governance of Layer 2 solutions, encouraging innovation and rewarding meaningful ecosystem contributions.
- **Regulatory Perspectives and Challenges:** The complexity of regulating Layer 2 technologies was acknowledged, with emphasis on the substantial knowledge gap facing policymakers. The importance of collaborative efforts to bridge this gap and develop informed regulatory approaches was highlighted.

Conclusions and Future Directions

The roundtable illuminated the dynamic and complex landscape of Layer Two governance, emphasizing the need for continuous innovation, collaboration, and regulatory engagement. Participants recognized the importance of forums that facilitate open dialogue among stakeholders, enabling a balanced approach to blockchain regulation that supports innovation while addressing potential risks. The discussions underscored the ongoing evolution of blockchain technologies and the collaborative effort required to navigate the challenges and opportunities they present.

Transcript (Anonymised) :

At a recent roundtable, Speaker 1 opened the session by stating it would operate under Chatham House rules, emphasizing confidentiality and non-attribution of the discussions. They highlighted the intention to utilize content provided by a contributor named Justin and engaged the audience for note-taking volunteers, stressing the importance of active participation.

Speaker 1 warmly invited everyone to contribute to the conversation, not just limiting it to the main panelists, aiming to foster a collaborative dialogue. The focus of the workshop was on governance in layer two blockchain solutions. Leona introduced herself and her project at Intermarks, focusing on developing a kinder, payment-oriented layer two solution to address transaction costs on Ethereum. She expressed eagerness to discuss the challenges within the layer two space.

Jack, a co-founder of Pier Two, detailed his company's involvement in the blockchain infrastructure realm since 2018, including running Ethereum validators and being significantly involved in the Arbitrum ecosystem. He mentioned the development of a light client in C sharp, indicating a broad engagement with blockchain technologies and ecosystems.

The conversation shifted towards the governance challenges encountered in developing zk (zero-knowledge proof) layer two solutions, particularly those mentioned by Justin. Leona highlighted a major issue: the complexity of zk proofs (ZKP) and the difficulty in auditing them, which represents a significant security challenge. She mentioned the potential of AI models, specifically LLMs, to automatically detect bugs in these protocols, emphasizing the rapid advancement of technology and its implications on human capability to maintain system security.

Leona elaborated on the main strategies for addressing these challenges, including the implementation of multi-signature and two-factor authentication methods to ensure the redundancy and integrity of the systems. She described this approach as mainstream, aiming to allow users to securely manage their assets even in the event of a security breach. The goal is to maintain the reliability of layer two solutions by ensuring that even if one part of the system is compromised, another remains secure, thus protecting users' assets.

This portion of the discussion underlines the ongoing challenges in developing secure, scalable blockchain solutions and the innovative approaches being explored to address these issues, including leveraging AI for security and developing robust governance models.

The conversation transitions into a detailed exploration of the challenges and potential solutions in securing zk (zero-knowledge proof) layer two technologies. Speaker 1 describes the adoption of multiple-signature (multisig) and two-factor authentication (2FA) as mainstream methods to address these challenges, alongside using multiple zkPs (zero-knowledge proofs) to ensure that if one system is compromised, another remains secure to protect user assets.

Speaker 1 raises a concern about the potential for automated systems, like AI models, to simultaneously exploit vulnerabilities in multiple security layers, posing new challenges in determining the optimal number of security layers needed for robust protection.

The discussion then pivots to the implications of adjusting penalty mechanisms for blockchain validators, specifically capping slashing penalties. Jack elaborates on his interest in the topic, particularly in the context of staking service providers. He underscores the disastrous consequences of significant slashing penalties and suggests that introducing a cap might not significantly affect staking service providers but could benefit other stakeholders by simplifying risk assessment for insurance purposes.

Jack further discusses the Eigen layer's introduction, allowing validators to repurpose staked Ethereum (ETH) for other applications, thus increasing the risk profile for validators. The conversation touches on the importance of establishing clear risk parameters for insurance and governance decisions, particularly in light of the potential for increased slashing events associated with the Eigen layer.

Speaker 1 introduces the concept of permissionlessness in the Eigen layer ecosystem, where the core team does not dictate the rules for application-specific blockchains (ABSs) and their slashing mechanisms. The discussion reflects on the theoretical and soon-to-be-implemented aspects of the Eigen layer, acknowledging that while initial ABSs may not introduce slashing mechanisms, future developments could bring additional risks for validators.

The dialogue concludes with a discussion on enshrining principles within layer two governance to ensure sustainability and mitigate risks, especially considering the advancing capabilities of AI models to detect or exploit vulnerabilities. The potential of integrating zk EVM (Ethereum Virtual Machine) into layer one for enhanced verification speed and scalability is also contemplated, highlighting the benefits of zero-knowledge proofs in securing and streamlining blockchain transactions across multiple layers.

This portion of the discussion delves into the future of layer two (L2) governance, particularly in the context of zero-knowledge (ZK) Ethereum Virtual Machines (EVM) and the potential for integrating such technologies into Ethereum's layer one (L1) to enhance scalability and verification processes. The conversation begins with Speaker 1 asking Leona about the principles she believes should be enshrined to ensure the sustainability of ZK layer two solutions amid the increasing power of AI models to identify or exploit vulnerabilities in ZK protocols.

Leona responds by highlighting the broad implications of incorporating ZK EVM directly into layer one, suggesting it could significantly reduce the verification time and costs associated with processing blockchain transactions. She raises a thought-provoking question: if ZK EVMs offer

such substantial benefits, why not implement them at layer one to directly enhance Ethereum's scalability and verification efficiency?

The discussion continues with Leona acknowledging that while integrating ZK EVM into layer one might reduce the perceived role of layer two solutions, it wouldn't necessarily diminish the importance of L2 governance. Instead, it prompts a reevaluation of L2's value proposition, particularly if core functionalities, such as scalability enhancements traditionally attributed to L2, are absorbed by layer one advancements.

Speaker 1 further explores the implications of such integration for layer two ecosystems, speculating on how it could shift the focus of L2 developments towards more specialized or customized solutions that go beyond merely enhancing scalability. Leona suggests that layer two could concentrate on advancing other aspects of blockchain technology, such as privacy enhancements or the development of alternative virtual machines (VMs) tailored to specific use cases, thereby contributing to a more diversified and resilient Ethereum ecosystem.

The conversation shifts towards the concept of retroactive public goods funding, a model that rewards contributors to the ecosystem based on their past contributions rather than potential future returns. This model is presented as a potential strategy for supporting the development and governance of L2 solutions, encouraging innovation and rewarding those who have meaningfully contributed to the ecosystem's growth.

In the concluding remarks, Speaker 1 reflects on the evolution of layer two protocols, tracing back to the inception of the Lightning Network for Bitcoin as an early example of L2 solutions aimed at increasing transaction throughput. This historical perspective underscores the ongoing evolution and diversification of L2 technologies, prompting a discussion about the collective understanding of what constitutes a layer two protocol and how this definition might continue to evolve in response to emerging challenges and opportunities in blockchain technology.

Throughout this segment, the dialogue underscores the dynamic interplay between layer one and layer two technologies, the potential for innovation within the Ethereum ecosystem, and the importance of governance models that adapt to technological advancements and shifting ecosystem needs.

The latter part of the discussion explores the intricate dynamics of layer two (L2) protocols, focusing on the intersection of open source initiatives, public goods, security models, and governance mechanisms. Speaker 1 contemplates the challenge of creating successful open source projects within the L2 space, acknowledging the significant contributions of historical open source initiatives like Apache, Nginx, Linux, and React JS. The discussion underscores the difficulty in adequately rewarding contributors in the open source domain, compared to the more substantial valuations and rewards seen in corporate settings.

The conversation shifts to a fundamental question regarding the definition and understanding of L2 protocols. Speaker 1 revisits the origin of L2 solutions, tracing back to the Lightning Network's introduction for Bitcoin as a method to increase transaction throughput. This historical perspective sets the stage for examining the evolving definition of L2 protocols, initially centered around scalability enhancements but expanding to include functionalities like zero-knowledge (ZK) proofs and permissioning systems.

A critical distinction is made between L2 solutions that inherit security assumptions from L1 protocols and those that introduce new security models or rely on semi-trusted parties, which may represent points of vulnerability. Speaker 1 argues that true L2 protocols must maintain the security integrity of L1 to be considered genuine extensions of the base layer.

The discussion delves into the nuances of governance within L2 ecosystems, contrasting it with L1 governance mechanisms. Speaker 1 elucidates the predominantly on-chain nature of L2 governance, facilitated by smart contracts and token-based voting systems. This governance model, while enabling rapid innovation and adaptation, introduces its own set of challenges, particularly in maintaining alignment with the security principles of L1 protocols.

Speaker 1 proposes that the pursuit of bug-free code from day one and the minimization or elimination of governance could theoretically align L2 security more closely with L1. However, the practicality of such approaches remains debatable, especially in the context of continuous development and the inherent complexities of blockchain technologies.

In conclusion, the discussion acknowledges the distinct yet interconnected roles of L1 and L2 protocols within the blockchain ecosystem. While L2 solutions offer a platform for experimentation and innovation beyond the more conservative governance and security frameworks of L1, ensuring the integrity and reliability of these layered solutions remains a paramount concern for developers, operators, and users alike.

This portion of the discussion delves into complex topics around layer two (L2) governance, the necessity and implications of using tokens within L2 protocols, and the broader question of what defines an L2 and its relationship to layer one (L1) security.

Speaker 1 reflects on Justin's insights regarding the inseparability of L2 governance from token mechanisms, given the need for on-chain protocol management. The conversation then shifts to pondering the feasibility of a tokenless L2, exploring the economic and social challenges such an approach would face, especially regarding incentivizing development and governance participation.

The dialogue explores the concept of shared sequencers and the potential conflicts between maximizing Ethereum's value versus preserving the sovereignty of individual L2 protocols. The participants discuss the alignment of interests among various stakeholders within the Ethereum ecosystem and the challenges posed by differing visions and goals.

Speaker 3, identifying as a representative from Polygon Labs, contributes their perspective on navigating governance without a native token, highlighting the challenges of rewarding contributions and coordinating among diverse stakeholders.

Speaker 1 further elaborates on the social complexities underlying tokenless rollups, emphasizing the difficulty of aligning various "tribes" within the crypto space, each with their own developers, communities, and venture capital backers. The discussion acknowledges that while technical solutions are crucial, social dynamics and stakeholder alignment pose significant challenges to governance and the successful implementation of L2 solutions.

The conversation then turns to the nature of permissionless design in L2 protocols and the essential characteristic of L2s: inheriting the security model of their corresponding L1 without negatively impacting it. This leads to discussions on the criteria and mechanisms for evaluating whether an L2 protocol genuinely upholds the security guarantees of its L1 counterpart.

As the dialogue unfolds, participants grapple with the practicalities of defining and recognizing true L2 protocols amidst a landscape of varied implementations and security assumptions. The discussion acknowledges the role of community consensus, expert evaluations, and potentially, dedicated organizations or initiatives aimed at clarifying and classifying L2 solutions based on their adherence to core security principles.

In conclusion, the discussion highlights the intricate balance between technical innovation, governance mechanisms, and the social dynamics that shape the development and adoption of L2 technologies within the broader Ethereum ecosystem. The conversation underscores the ongoing need for collaboration, rigorous analysis, and community-driven approaches to navigate the evolving landscape of blockchain governance and scalability solutions.

The conversation wraps up by reflecting on the complexities and challenges associated with regulating layer two (L2) technologies within the blockchain space. Speaker 1 highlights the pivotal role of initiatives like L2 Beat, which, with Ethereum Foundation's support, provides clarity on the operational and security aspects of various L2 solutions. The dialogue acknowledges the importance of such efforts in informing regulatory and community perspectives on emerging technologies.

Discussion touches upon the regulatory landscape's evolution and the substantial lag between technological advancements and the formulation of corresponding regulatory frameworks. The conversation underscores the significance of regulatory awareness and understanding of permissionless innovation on L2 platforms, juxtaposing it with the decade-long efforts to comprehend and legislate around L1 technologies.

The dialogue then delves into the potential regulatory concerns surrounding privacy-preserving L2 solutions. Participants ponder the implications of such technologies for illicit activities, questioning the responsibility and capacity of platforms like Ethereum to address these issues without compromising their core principles.

Contributions from various speakers, including representatives from regulatory bodies and industry insiders, illuminate the substantial knowledge gap facing policymakers. This gap hinders effective regulation of complex, rapidly evolving technologies like L2 solutions. Participants call for enhanced educational efforts to bridge this gap, enabling more informed regulatory approaches that can adapt to the nuanced landscape of blockchain technologies.

As the discussion unfolds, it becomes evident that defining and regulating L2 technologies poses unique challenges, not only due to the technical intricacies involved but also because of the diverse interpretations and applications of L2 concepts across the blockchain ecosystem. The conversation highlights the necessity for a collaborative approach involving technical experts, regulators, and the broader community to develop coherent, flexible regulatory frameworks that accommodate the innovative potential of L2 solutions while addressing associated risks.

In concluding, the session emphasizes the ongoing need for dialogue and education among stakeholders to navigate the complex interface between blockchain innovation and regulatory frameworks. The importance of forums that facilitate such exchanges is acknowledged, alongside a collective commitment to fostering a balanced, forward-looking approach to blockchain regulation that supports innovation while safeguarding against potential harms.